Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/16/2017
PI Name:	Poirier, David Sc.D.		
Project Title:	Effect of Varying Convection on Dendrite Morphology and Macrosegregation		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2014	**End Date:**	11/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2016 report: Prof. D.R. Poirier, remains as the P.I. Dr. V. Manga, Assistant Research Professor, in the Dept. of Materials Science and Engineering is now co-investigator at The University of Arizona. He replaced Dr. N. Swinteck who resigned the position in August, 2015 to take a position in industry.		
COI Name (Institution):	Tewari, Surendra (Cleveland State University) Manga, Venkat Ph.D. (The University of Arizona)		
Grant/Contract No.:	NNX15AC04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., 12/16/2014): This project is a continuation of grant NNX10AV40G, "Effect of Varying Convection on Dendrite Morphology and Macrosegregation," with period of performance of 12/1/2010-11/30/2014. The research of this grant pertains to convection that is driven by directional solidification (DS) through changes in cross-sections (because of solidification shrinkage) and thermosolutal convection. In microgravity, of course, the thermosolutal convection will be mitigated so that effects of the shrinkage-driven flow in cross-sectional changes can be isolated from the usual thermosolutal convection that always occurs when solidification occurs on Earth. The outcome of the research is expected to be relevant to commercial DS-castings, that are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and electric power generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research not only has scientific merit, it is also relevant to the production of commercial DS castings. The research comprises the following: 1) directional solidification experiments that are compatible with the solidification cartridge assemblies (SCAs) that can be employed in a furnace on the International Space Station (ISS). The research comprises: 1) selected alloys have varying density-responses to gravity forces; 2) crucible designs with variable cross-sections but compatible with the current SCAs; 3) sophisticated characterizations of the solidified microstructures, including both micro- and macro-segregation; and 4) supporting computer simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The outcome of the research is expected to be relevant to commercial directionally solidified-castings, that are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and turbine electric power-generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research is very much needed for its scientific merit and also because it is relevant to the production of commercial DS castings.
Task Progress:	Experiments and computer simulations of the solidification of two hypoeutectic alloys were accomplished in previous reporting periods. The subject alloys were selected because they can result in a type of macrosegregation, which is known "steepling." For example, in the case of the Al-7 wt.% alloy when it undergoes steepling, the central portion of the casting is only 5 wt.% Si whereas the periphery is 12.6 wt.% Si. When the solidification occurs through mold-contractions and/or mold-expansions, complex convection patterns further complicate the macrosegregation patterns. During this reporting period, we have embarked on simulating the solidification of a binary alloy (Pb-6Sb) with very different characteristics of the two Al-alloys. The macrosegregation in this alloy is known as "freckling." Experiments were done at Cleveland State University. At the University of Arizona (UA) we have concentrated on collecting the solidification properties and other properties of the alloy in order to effect the computer simulations. In addition to doing the simulations using the multi-shared supercomputer at UA, our NASA Project Engineer has helped us to gain access to the NASA supercomputer at AMES. This will enable to carry out the simulations much faster. Because the properties of the Pb-Sb alloy differ so much from the two Al-alloys, we expect that the convection and resulting macrosegregation patterns will be very different than previously seen in the Al-alloys.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Ghods M, Lauer M, Grugel RN, Tewari SN, Poirier DR. "Convection and macrosegregation in Al-19Cu alloy directionally solidified through an abrupt contraction in cross-section: A comparison with Al-7Si." Journal of Crystal Growth. 2017 Feb 1;459:135-45. http://dx.doi.org/10.1016/j.jcrysgro.2016.11.111 , Feb-2017
Articles in Peer-reviewed Journals	Ghods M, Johnson L, Lauer M, Grugel RN, Tewari SN, Poirier DR. "Macrosegregation in Al-7Si alloy caused by abrupt cross-section change during directional solidification." Journal of Crystal Growth. 2016 Sep;449:134-47. http://dx.doi.org/10.1016/j.jcrysgro.2016.06.010 , Sep-2016